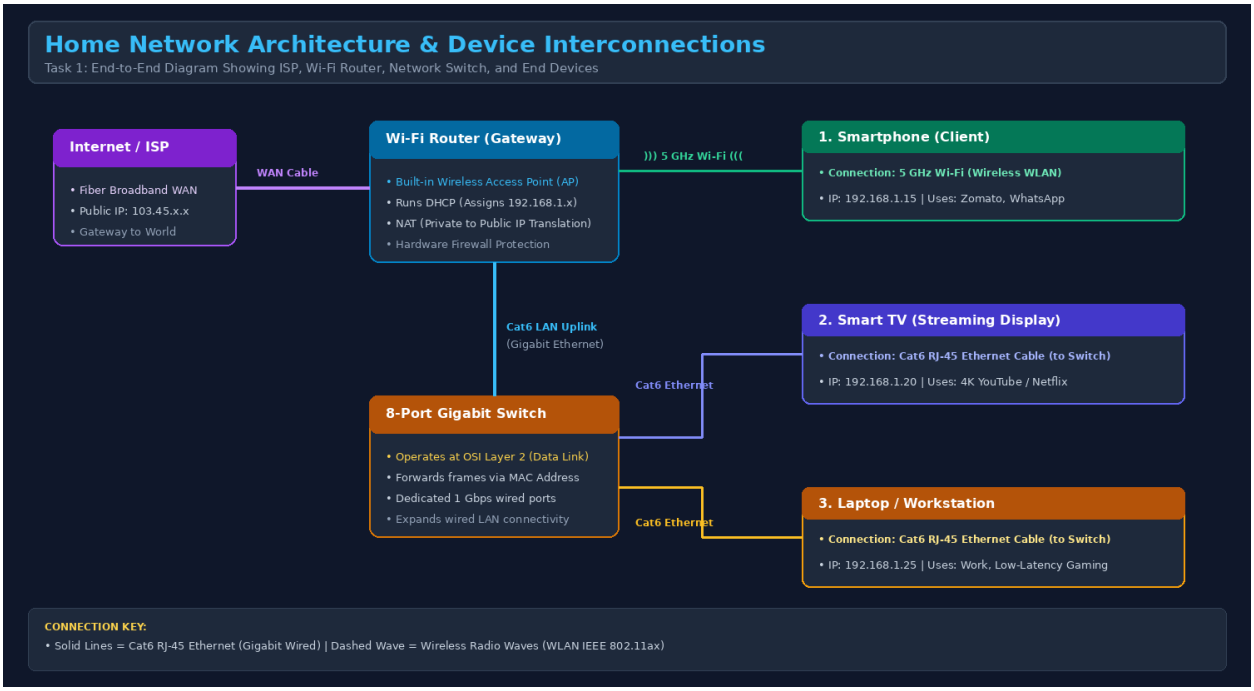
Network Devices

Task 1: Home Network Architecture Diagram

The diagram below maps the complete path from the Internet Service Provider down to your local end devices:



Device Connection Summary:

1. **Internet (ISP) →\rightarrow→ Wi-Fi Router:** Connected via an optical fiber patch cord or Ethernet cable into the router's **WAN port**.
2. **Wi-Fi Router →\rightarrow→ Smartphone:** Connected wirelessly over **5 GHz5\text{ GHz}5 GHz Wi-Fi** using the router's built-in Access Point (WLAN).
3. **Wi-Fi Router →\rightarrow→ Gigabit Switch:** Connected with a **Cat6 Ethernet cable** running from a router LAN port down to Port 1 of the network switch (LAN uplink).
4. **Switch →\rightarrow→ Smart TV & Laptop:** Both devices connect directly into individual switch ports using **Cat6 RJ-45 Ethernet cables**, providing dedicated, interference-free wired speeds.

Task 2: Network Devices Comparison Table

Network Device	Operating OSI Layer	How It Forwards Data	Real-World Use Case & Daily App
Router	**Layer 3**	*(Network)* | Examines destination **IP addresses** to forward packets between different networks (connects your private home LAN to the public Internet WAN). Performs DHCP and NAT.	**Streaming 4K Videos on YouTube:** The router translates your device's private IP (192.168.1.15) into a public IP to request and fetch video stream chunks from Google's content servers.
Switch	**Layer 2**	*(Data Link)* | Examines destination **MAC addresses** stored in a CAM table to intelligently forward frames directly to the specific target port (dedicated full-duplex bandwidth).	**Low-Latency LAN Gaming (Valorant / CS2):** Multiple PCs connect through a gigabit switch to eliminate Wi-Fi packet drops, jitter, and ping spikes during competitive matches.
Hub	**Layer 1**	*(Physical)* | Repeats and broadcasts every incoming electrical signal out of **all other ports** blindly (creates a single shared collision domain).	**Legacy Network Traffic Sniffing:** Connecting a diagnostic laptop running Wireshark to mirror and inspect raw network packets across legacy industrial systems.
Wireless Access Point (AP)	**Layer 2**	*(Data Link)* | Bridges wireless 802.11 RF radio signals from mobile devices into standard 802.3 Ethernet frames on the wired LAN.	**Browsing Instagram Reels on Phone:** Wirelessly broadcasts 2.4 GHz/5 GHz2.4\text{ GHz} / 5\text{ GHz}2.4 GHz/5 GHz signals across the living room so mobile devices can access the high-speed wired network without physical cables.

Task 3: Data Path When Ordering Food on Zomato

When you tap **"Place Order"** in the **Zomato** app on your smartphone, data travels through the network in the following sequence:

1. **Smartphone →\rightarrow→ Wireless Access Point (AP):**

Your phone encapsulates your food order and payment session into encrypted Wi-Fi radio waves (802.11ax802.11\text{ax}802.11ax). The antenna of the **Access Point** (integrated inside your home router) receives these radio waves and translates them into physical Ethernet frames.

2. **Access Point →\rightarrow→ Network Switch:**

The AP passes the Ethernet frames across the internal **Switch**. The switch reads the destination MAC address to ensure the frame is cleanly directed to the router's internal gateway without collision.

3. **Switch →\rightarrow→ Router:**

The **Router** processes the packet at Layer 3. It strips the local header, applies **Network Address Translation (NAT)** to replace your phone's private IP (192.168.1.15) with your broadband subscription's public IP, and forwards the packet out through its WAN port into your ISP's fiber-optic infrastructure.

4. **Internet →\rightarrow→ Return Trip:**

The order reaches Zomato's cloud backend on AWS, processes the transaction, and sends back an order confirmation. The router receives the incoming packets, reverses NAT, passes the frames through the switch to the AP, and the AP broadcasts the signal back to your phone, updating your screen with *"Order Confirmed"*.

Task 4: Connecting an Ethernet-Only Gaming PC Far from the Router

To connect a gaming PC located far from the router without running long, messy cables across hallways, here are two recommended solutions:

Option 1: Powerline Ethernet Adapter Kit

- **How It Works:**

A Powerline kit consists of two plug-in wall adapters that turn your home's existing electrical copper wiring into high-speed network cables.

1. Plug **Adapter A** into an electrical wall outlet near your Wi-Fi router and connect it to a router LAN port using a short Cat6 cable.
2. Plug **Adapter B** into a wall socket next to your gaming PC and run an Ethernet cable directly from the adapter into your PC's LAN port.

- **Why It Is Ideal for Gaming:** It delivers a stable, physical wired link with consistently low ping and minimal packet jitter compared to wireless signals passing through walls.

Option 2: Wi-Fi Extender / Mesh Node with an Ethernet Port (Wireless Bridge Mode)

- **How It Works:**

A Wi-Fi range extender or Mesh satellite node is placed near your gaming desk.

1. The extender connects wirelessly to your main Wi-Fi router over high-speed 5 GHz5\text{ GHz}5 GHz or 6 GHz6\text{ GHz}6 GHz backhaul.
2. You plug a standard RJ-45 Ethernet cable from the **Ethernet LAN port on the extender** straight into your gaming PC.

- **Why It Works Here:** The extender acts as an external wireless bridge/client. Your gaming PC receives a stable wired Ethernet handshake, while the extender handles the wireless transmission back to the main router.